clock. It is true that the binding energies of nuclei are very small, but so are their masses too. Therefore, the mass $m = E/c^2$ is no longer negligibly small in comparison with the masses of the nuclei themselves. Consequently, any change taking place in the nuclei as a result of changes in energy content, can be measured with a considerable degree of accuracy. The application of the above formula to atomic nuclei confirms at the same time the important relationship it expresses.

Since energy is liberated when a deuteron is formed from a proton and a neutron, the mass of the deuteron must be less than the sum total of the masses of the proton and the neutron while these two particles still had a separate existence. Thus must apply to every nucleus composed of N neutrons and Z protons. This statement can be expressed in the form of an equation as follows:

$$M_{nucleus} = Nm_{neutron} + Zm_{proton} - \frac{|E|}{C^2}$$

where $|E|$ is the binding energy, taken as a positive magnitude, of the nucleus, which in the case of the deuteron is liberated in the form of a photon.

It is convenient to use the above equation in a slightly modified form, referring it to the neutral atom, i.e. to the nucleus plus the extranuclear atomic structure, as a whole, instead of the nucleus alone. In this case, the hydrogen atom of mass number 1, with its one electron, must be substituted for the proton. As a result, the masses appearing on both sides of equation increase by the mass of the Z electrons possessed by the entire atom, on the one hand, and by the Z protons, on the other. The equation then will read:

$$M_{atom} = Nm_{neutron} + Zm_{H\,atom} - \frac{|E|}{C^2}$$

From this equation we can compute the binding energy of a nucleus if we know exactly the mass of the atom in question, the mass of the neutron and the mass of the hydrogen atom.

The mass of an atom is thus always $|E|/c^2$ less than the sum of the masses of its component parts. This difference in mass is called the *mass defect* of the nucleus. It equals $|E|/c^2$, and therefore:

$$mass\ defect = \frac{|E|}{C^2} = Nm_{neutron} + Zm_{H\,atom} - M_{atom}$$

Atomic masses are usually expressed in terms of the standard physical